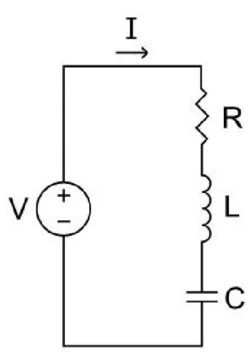
Name

$M\#$

Models

- Mixing in Tanks https://xkcd.com/2974/
  - $y' = A\,y + f$ with $y(0) = y_0$ given.
  - Conservation of volume v and salt s for each tank
    - $v'(t) =$ rate in $-$ rate out
    - $s'(t) =$ rate in $-$ rate out
- Probes in Earth-Moon system. CR3BP https://orbital-mechanics.space/the-n-body-problem/circular-restricted-three-body-problem.html
    - With $M = 1/82.45$ and $E = 1 - M$ and $r_1(x, y) = \sqrt{(x+M)^2 + y^2}$ and $r_2 = \sqrt{(x-E)^2 + y^2}$

$$\begin{aligned} x'' &= 2\,y' + x - \frac{E(x+M)}{r_1^3} - \frac{M(x-E)}{r_2^3} \\ y'' &= -2\,x' + y - \frac{E\,y}{r_1^3} - \frac{M\,y}{r_2^3} \end{aligned} \quad \text{with } y(0),\ x(0),\ x'(0),\ y'(0) \text{ given}$$

    - Follows from Newton's law of gravitation in a rotating coordinate system.
- RLC circuits https://en.wikipedia.org/wiki/RLC_circuit
  - Circuit Rules: https://en.wikipedia.org/wiki/Kirchhoff%27s_circuit_laws
    - Current i is the rate of change of charge q (usually on a capacitor) through a wire.
    - Voltage drops due to resistors ($V_R = i\,R$), capacitors ($V_c = \frac{q}{C}$), inductors ($V_l = L\,\mathrm{di}/\mathrm{dt}$), and voltage sources sum to zero around a circuit loop.
    - Currents sum to zero at a junction.
  - 

ODE

$V + q'R + q''L + q/C = 0$

q is charge on the capacitor

current $i = q'$ Labelled I in figure

Figure from wiki page above.

R, L, and C are constants

- Population models https://en.wikipedia.org/wiki/Population_model
  - Growth of population p
    - $p'(t) =$ birth rate $-$ death rate

- Predator (wolf w) prey (moose m) https://en.wikipedia.org/wiki/Lotka%E2%80%93Volterra_equations
  - $m'(t)$ = birth rate − death rate (Assume moose death rate proportional to wolf and moose population)
  - $w'(t)$ = birth rate − death rate (Assume wolf birth rate proportional to wolf and moose population)

```
{α, β, γ, δ} = {1.2, 2.3, 3.4, 5.6};
TMax = 12;
{w0, m0} = {4.3, 2.3};
{mSol, wSol} = NDSolveValue[{
   m'[t] == α m[t] - β m[t] * w[t], m[0] == m0,
   w'[t] == -γ w[t] + δ m[t] * w[t], w[0] == w0
   }, {m, w}, {t, 0, TMax}]

TabView[{
  "pop vs t" → Plot[{mSol[t], wSol[t]},
    {t, 0, TMax}, AxesLabel → {"t", "pop"}, PlotLegends → {"m", "w"}],
  "pop vs t" → LogPlot[{mSol[t], wSol[t]},
    {t, 0, TMax}, AxesLabel → {"t", "pop"}, PlotLegends → {"m", "w"}],
  "m vs w" → ParametricPlot[{mSol[t], wSol[t]},
    {t, 0, TMax}, AxesLabel → {"m", "w"}, PlotRange → All]
 }]
```

First order ODE: Analytical Solutions

- A <u>separable</u> scalar ODE has the form $y'(t) = g(t)\, h(y)$.

  You solved these in Calculus II by separating variables and integrating

  $$dy/dt = g(t)\, h(y) \Longrightarrow \int \frac{dy}{h(y)} = \int g(t)\, dt + C$$

  An initial condition $y(0) = y_0$ would let you solve for the one constant of integration.
- A <u>linear</u> 1st order ODE has the standard form $y'(t) + p(t)\, y(t) = f(t)$. Solve by computing the IF $v(t) = e^{\int_0^t p(\tau)\, d\tau}$ and solve the separable ODE $(v\, y)' = v\, f$.
- A 1st order ODE of the form $\frac{\partial f}{\partial t}\, dt + \frac{\partial f}{\partial y}\, dy = 0$ is <u>exact</u> with general solution $f(t, y) = C$.
  - $M(t, y)\, dt + N(t, y)\, dy = 0$ is exact if and only if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial t}$
  - To solve compute $f(t, y) = \int M(t, y)\, dt + h(y) \Longrightarrow h'(y) = N(t, y) - \frac{\partial}{\partial y}\left[\int M(t, y)\, dt\right]$ then integrate $h'(y)$ and write down the general solution $f(t, y) = C$
- 2.7 Substitution. Some ODEs look nicer in other variables. For example
  - 2.7.4 Bernoulli: $y' + p(t)\, y = q(t)\, y^n$. The substitution $u = y^{-n}$ gives a linear equation in u.